

# Ladle TAT Optimisation — Thermal × Kinetic × Scheduling Framework

Reducing IF → LF → Caster turnaround from ~70 min to ≤ 40 min via two-stage dephosphorisation and queue-aware LF scheduling

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## TARGET & HEADLINE RESULT

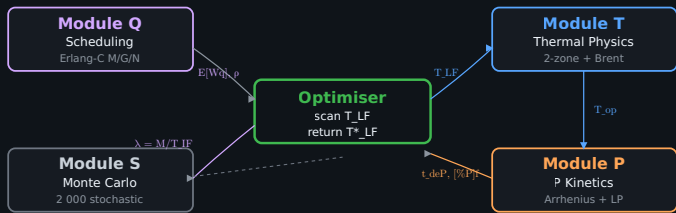
TAT target: **30–40 min** · current: **~70 min**

✓ **Case 1 (8 IF + 2 LF) — deterministic TAT 37.6 min · MC P(TAT≤40) = 91.5%**

Arc energy **27 kWh/T** vs 100 kWh/T limit · 2.4 min headroom

## \$2 Four-Module Architecture

tightly coupled — not sequential



## \$\$4–6 Governing Equations

**Q** Erlang-C queue wait — multi-server LF system

$$C(N, \rho) = \frac{(N\rho)^N / [N!(1-\rho)]}{\sum_{n=0}^{N-1} (N\rho)^n / n! + (N\rho)^N / [N!(1-\rho)]} \quad E[W_q] = C/(N\mu - \lambda)$$

**P** Temperature-coupled deP rate constant

$$k_{LF}(T) = k_{base} \cdot \exp[-E_a/R \cdot (1/T - 1/T_{ref})] \cdot \exp[3000 \cdot (1/T - 1/T_{ref})]$$
$$E_a = 50\,000 \text{ J/mol} \cdot T_{ref} = 1843 \text{ K (1 570 °C)} \cdot k_{base} = 0.0116 \text{ min}^{-1} \cdot \textit{peaks at 1 570 °C}$$

**P** Slag-chemistry kinetic enhancement (Turkdogan · Mazumdar–Evans)

$$k_{LF}^{enh} = k_{base} \cdot (B_{new}/B_{old})^{0.5} \cdot (Q_{new}/Q_{old})^{0.4} \cdot ([\%FeO]_{new}/[\%FeO]_{old})^{0.5} = 4.43 \times k_{base}$$

**T** Two-zone cooling + Brent back-solve for  $T_{exit,LF}$

$$dT/dt|_{steel} = (\dot{Q}_{rad} + \dot{Q}_{cond}) / (m_s C_{p,s}) \quad \text{Find } T_{exit}: \int_0^{t_{tr2}} dT/dt \, dt = T_{cast} - T_{exit}$$

**Q** TAT budget decomposition · binding constraint

$$TAT = t_{tap} + t_{tr1} + E[W_q] + T_{LF} + t_{tr2} \leq 40 \text{ min} \Rightarrow E[W_q](T_{LF}) + T_{LF} \leq 15 \text{ min}$$

## \$3 · CENTRAL RESULT · MATHEMATICAL NECESSITY

### Single-Stage LF Dephosphorisation is Physically Impossible

From first-order kinetics:  $k_{LF}^{req} = \ln([\%P]_o/0.040)/T_{LF}$ . With  $T_{LF} = 12$  min and full slag enhancement (basicity 2.5 → 4.5, Ar 0.1 → 0.4 L/min/T, FeO 5 → 18%), the **maximum physically achievable** rate constant is  **$k_{LF}^{max} \approx 4.43 \times 0.0116 = 0.0514 \text{ min}^{-1}$** . In every realistic scenario,  $k^{req}$  exceeds this ceiling — no slag chemistry can close the gap.

**Two-stage processing is therefore not a design preference. It is a mathematical necessity.**

| SCENARIO                                       | [%P] <sub>o</sub> | $k^{REQ}_{LF}$ (MIN <sup>-1</sup> ) | SHORTFALL | OK? |
|------------------------------------------------|-------------------|-------------------------------------|-----------|-----|
| Minimum input                                  | 0.080 %           | <b>0.0578</b>                       | −12 %     | ✗   |
| Nominal input                                  | 0.090 %           | <b>0.0676</b>                       | −31 %     | ✗   |
| Worst case                                     | 0.100 %           | <b>0.0764</b>                       | −49 %     | ✗   |
| <b><math>k^{max}</math> (full enhancement)</b> |                   | <b>0.0514</b>                       | ceiling   |     |

### ✓ SOLUTION · TWO-STAGE

Exploit the IF residence time (150–180 min) for **40 min IF pre-treatment** at  $k_{IF} = 0.010 \text{ min}^{-1}$ , then LF polish at  $k^{enh} = 0.0514 \text{ min}^{-1}$ .

**Stage 1 (IF, 40 min): 0.100% → 0.0670%**

**Stage 2 (LF, ≈10 min): 0.0670% → 0.040% ✓**

## \$4.3

### 5-COMPONENT

### ARC ENERGY

#### [1] STEEL SENSIBLE

$$m_s \cdot C_p \cdot \Delta T_{arc}$$

5–15 kWh/T

≈ **8.0 kWh/T**

#### [2] SLAG HEATING

$$4\% \text{ mass} \cdot \Delta T \, 200 \text{ °C}$$

2–5 kWh/T

≈ **2.7 kWh/T**

#### [3] REFRACTORY

$$E_{ref,nominal} \text{ (fixed)}$$

8–15 kWh/T

**8.0 kWh/T**

#### [4] DYNAMIC LOSSES

$$\int \dot{Q}_{rad+cond} dt$$

3–8 kWh/T

≈ **3.5 kWh/T**

#### [5] DeP EXOTHERM

$$-n_p \cdot \Delta H_{deP} / 3600$$

credit

≈ **−2.1 kWh/T**

$$E_{LF} = \Sigma[1..5] / \eta$$

$$\approx \mathbf{27 \text{ kWh/T}}$$

η = 0.75 (normal slag) · limit

100 kWh/T

### ● Case 1 · 8 IF + 2 LF

#### ✓ FEASIBLE

#### TOTAL TAT

**37.6** min

headroom 2.4 min

#### T\*<sub>LF</sub> OPTIMAL

**11.6** min

feasible 11.6–15.3

### ● Case 2 · 4 IF + 1 LF

#### ✗ INFEASIBLE

#### BEST TAT (DET.)

**41.1** min

+1.1 min over limit

#### MC P(TAT ≤ 40)

**89.2** %

216 / 2 000 breach

### ● Monte Carlo Inputs & Scaling Rule

$X_i \in \{m, [\%P]_o, \bar{T}_B, S\}$  drawn independently per heat ·  $n = 2\,000$

#### Arrivals

Erlang-3 (k=3, CV=1/√3)

#### Service time

Log-Normal, CV = 0.20

#### Heat tonnage $m$

U[13, 17] T

#### [%P]<sub>init</sub>

U[0.08, 0.10] %

#### Zone B cooling rate

U[2.0, 3.0] °C/min

## \$7.3 · GENERAL M IF + N LF SCALING

$$N_{min} = \lceil M \cdot T_{LF}^* / [T_{IF}(1 - \rho_{max})] \rceil = \lceil M / 8.25 \rceil$$

At  $T_{IF} = 165$  min,  $T_{LF} \approx 12$  min,  $\rho_{max} = 0.40$  advisory. **+1 LF per ~8 additional IFs** to keep TAT ≤ 40 min.



|                              |               |            |
|------------------------------|---------------|------------|
| Utilisation ρ                | 0.281         | ✓<br><0.40 |
| E[W <sub>q</sub> ]           | 0.99 min      | ✓<br><15   |
| T <sub>at</sub> at caster    | 1 625 °C      | ✓<br>MET   |
| Arc energy (15 T / 13 T)     | 22 / 32 kWh/T | ✓<br><100  |
| [%P] <sub>f</sub> worst case | 0.040 %       | ✓<br>MET   |

#### MONTE CARLO · 2 000 HEATS

Mean / P95 TAT = **36.6 / 40.9 min** · **P(TAT ≤ 40) = 91.5%** · 0 violations on energy, %P, T<sub>cast</sub>

|                           |          |                  |
|---------------------------|----------|------------------|
| Utilisation ρ             | 0.281    | ✓<br>stable      |
| E[W <sub>q</sub> ]        | 4.52 min | ✓<br><15         |
| T <sub>at</sub> at caster |          | ✗ TAT constraint |
| Arc energy (15 T)         | 29 kWh/T | ✓<br><100        |
| [%P] worst case           | 0.040 %  | ✓<br>MET         |

#### ROOT CAUSE

No  $T_{LF}$  satisfies all 6 constraints simultaneously. Thermal & chemistry OK — **queue-driven delay is binding**. Fix: add a 2nd LF → Case 1.

### ⚡ BOTTLENECK

**Scheduling — not arc power — binds**. LF hardware is thermally sufficient; 70-min cycle driven by sub-optimal slag chemistry, not physics.

### 📏 SCALING LAW

$N_{min} = \lceil M/8.25 \rceil$  LFs for M IFs. Stepwise expansion in discrete LF tiers. Validated across M = 1...40, N = 1...5.